Sim-to-Real Characterization of a 1-DOF Antagonistic Tendon-Driven Joint

H0 Experimental Platform — Technical Project Report

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Repository: https://github.com/thezeta/antagonistic-joint-sim2real

1. Introduction

Tendon-driven and musculoskeletal robotic systems offer a way to reproduce some of the compliance and antagonistic actuation found in biological mechanisms. Unlike rigid transmissions, however, the relationship between actuator motion and joint motion is influenced by tendon elasticity, preload, routing geometry, friction, and other non-ideal effects. These properties make such systems attractive for compliant actuation, but they also make their behavior more difficult to predict accurately using simplified physical models.

This project investigates that gap using a minimal one-degree-of-freedom antagonistic tendon-driven joint. The system consists of two independently actuated tendons connected to opposing sides of a rotary joint through extension springs. Motor-side incremental encoders provide actuator-position feedback, while an absolute magnetic encoder measures the resulting joint angle. A physics-based simulation of the same mechanism is used to predict the joint configuration resulting from commanded tendon displacement.

The purpose of the project is not initially to eliminate every source of mismatch between simulation and hardware, but to construct and commission a sufficiently controlled physical platform that the mismatch itself can be measured meaningfully. Particular attention is therefore given to repeatable initialization, actuator positioning, tendon preload, and experimental characterization before performing matched simulation and hardware trials.

The resulting sim-to-real discrepancy is first treated as an experimental observation and then used to motivate progressively richer compact models. The study therefore examines not only whether the nominal physics-derived model departs from the physical system, but also which experimentally observed forms of memory and relaxation are necessary to explain and predict that departure.

2. System Design

The experimental platform is a one-degree-of-freedom antagonistic tendon-driven joint developed as a minimal physical system for studying the relationship between actuator displacement and joint configuration. The joint rotates a rigid arm in the horizontal plane about a vertical shaft and is actuated by two opposing tendon paths.

Each tendon is driven by a Namiki 22CL-3501PG geared brushed DC motor through a dedicated winding spool. The two actuators operate antagonistically: differential winding changes the relative tendon lengths and produces rotation of the joint. An extension spring is incorporated into each tendon path to provide compliance and maintain tendon tension. The springs have a free length of 50 mm. The canonical experimental condition uses a standardized common preload of +100 winding counts on both motors, with all test coordinates expressed as differential offsets about that preload.

Motor actuation is controlled by an STM32 NUCLEO-F446RE through two BTS7960B motor drivers. Incremental encoders mounted on the motor side provide winding-position feedback to a damped encoder-feedback servo. The resulting joint configuration is measured independently using an AS5600 absolute magnetic encoder located at the rotary joint.

The electrical system is powered from a 12 V, 5 A supply. The motor-driver power stage and microcontroller logic share a common ground, while the breadboard is used only for low-current logic, sensor distribution, and encoder pull-ups. For normal experimental operation, the control electronics and associated wiring are placed beneath a removable 3D-printed cover. The cover has no mechanical or electrical function in the experiment; it protects the electronics and keeps the visual field clear during recorded runs.

The separation between motor-side and joint-side sensing is central to the experimental design. The motor encoders describe the realized actuator state, while the AS5600 independently measures the resulting joint angle. This makes it possible to distinguish actuator-positioning error from discrepancies arising within the compliant tendon transmission and to compare a model prediction based on realized actuator state with the configuration produced by the physical mechanism.

The complete mechanical assembly is maintained in the project repository at:
/hardware/cad/h0_antagonistic_joint_assembly.stp.

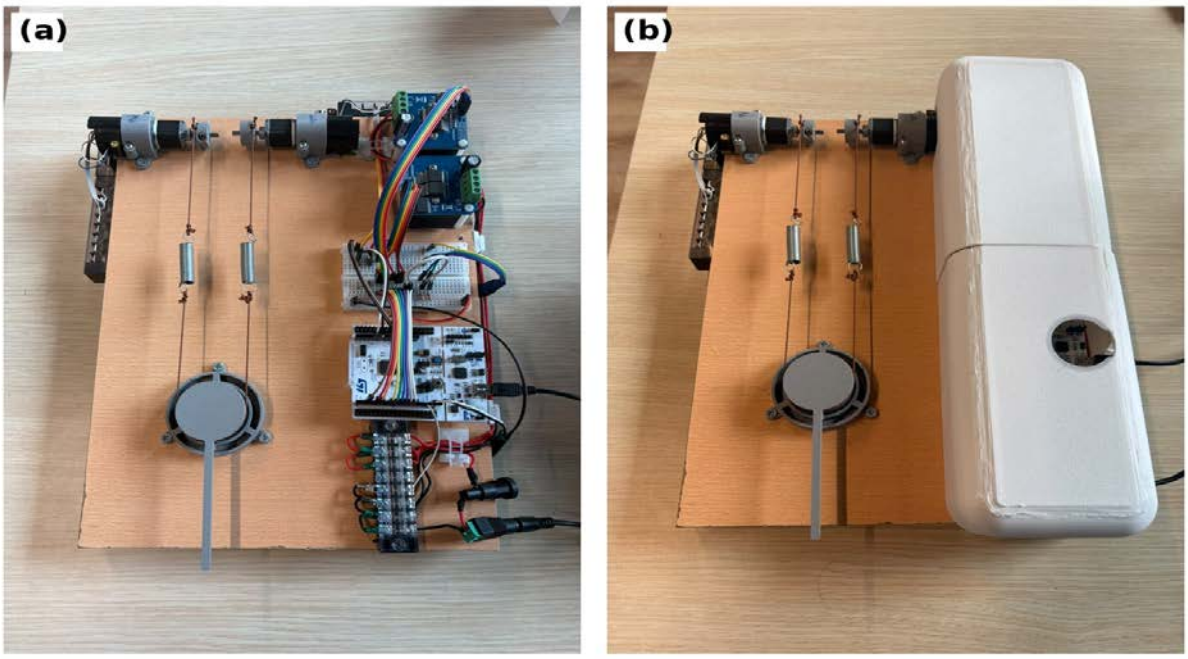


Figure 1. H0 experimental platform: (a) electronics exposed for hardware documentation; (b) normal covered configuration used for clean experimental recording.

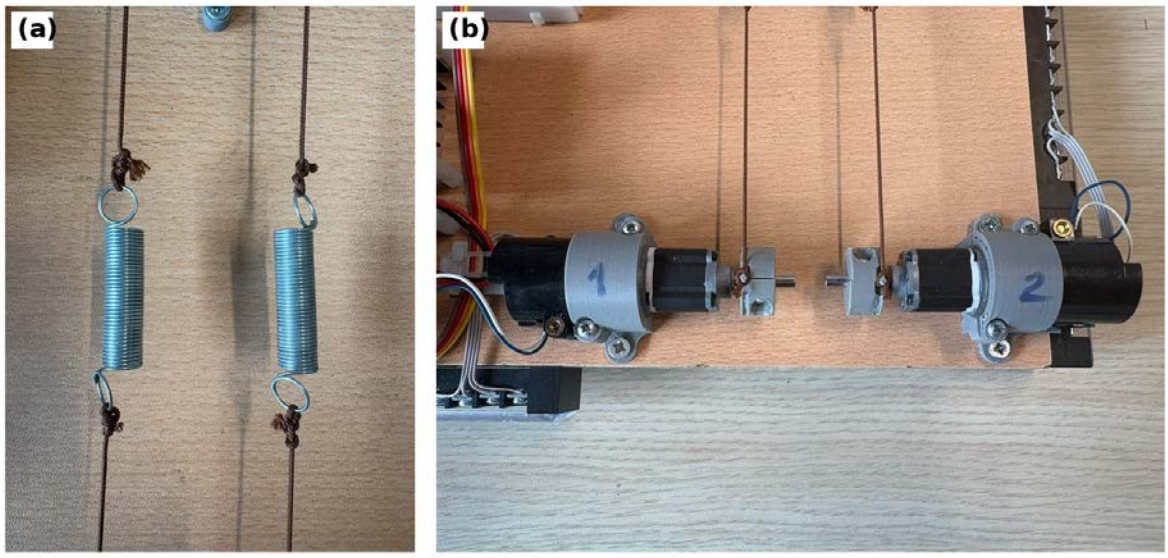


Figure 2. Mechanical transmission details: (a) extension-spring tendon elements; (b) geared motors, winding spools, and tendon attachment geometry.

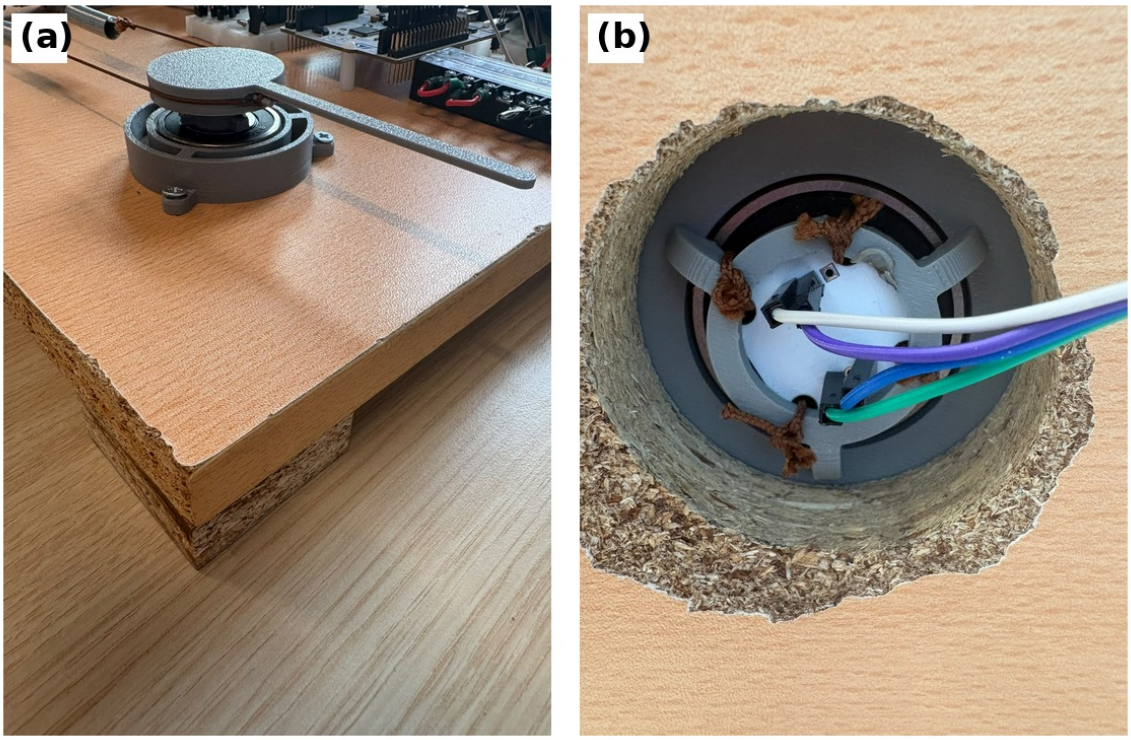


Figure 3. Joint and sensing details: (a) side view of the rotary joint and output arm; (b) underside integration of the absolute magnetic joint-angle sensor.

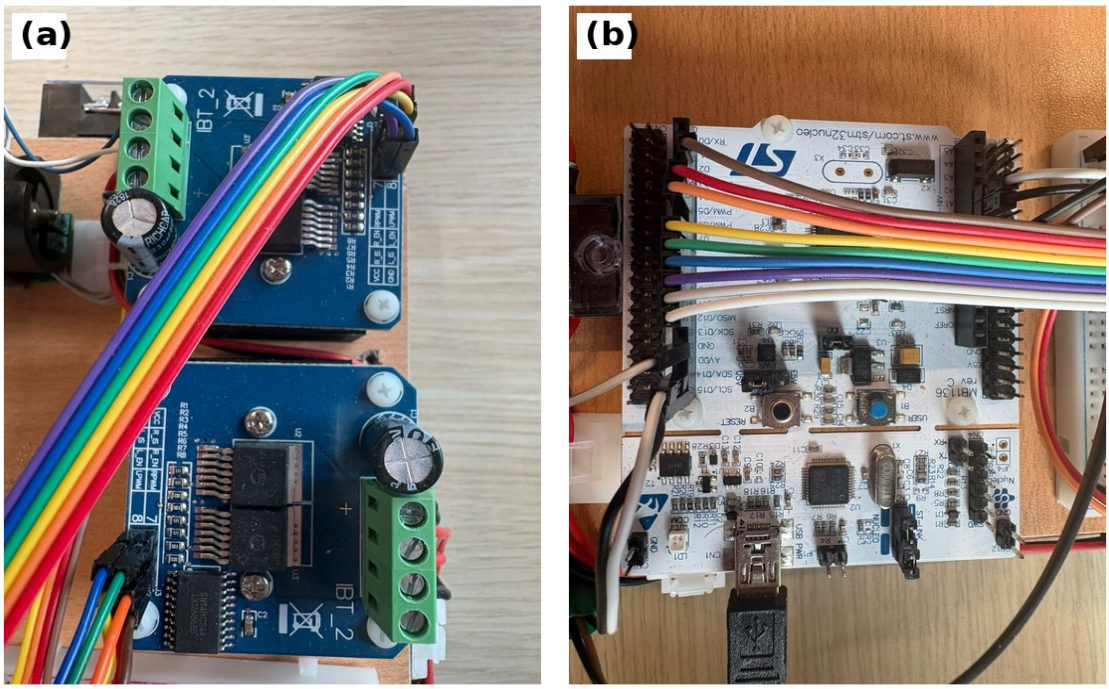


Figure 4. Control electronics: (a) dual BTS7960 motor-driver modules; (b) STM32 NUCLEO-F446RE controller and signal wiring.

A move demonstration video showing three Linux terminal move commands and the corresponding H0 physical responses can be found at **/media/move_demo.mp4** and in **README**.

3. Model and Simulation

A physics-based simulation of the antagonistic joint was developed before the physical platform was completed. Its purpose is to provide a deliberately simplified prediction of joint configuration from actuator-side tendon displacement so that the behavior of the real mechanism can be compared against a known model.

The simulated system reproduces the principal geometry of the physical plant: two antagonistic tendon paths act on opposite sides of a single rotary joint, with one extension spring in each tendon path. Motor rotation is converted into tendon-length change through the winding spools, and the resulting spring extensions generate opposing tendon forces. In the quasi-static limit, the equilibrium joint configuration is obtained from the balance of the antagonistic transmission geometry.

3.1 Model v0: Nominal Physics-Derived Model

Model v0 is a memoryless physics-derived representation of the antagonistic transmission. In the quasi-static limit, the joint angle is approximately

$$q = \frac{(w_2 - w_1)}{(2\, r_j)}$$

Here w_1 and w_2 denote tendon-length changes and r_j is the effective joint radius. Using the calibrated tendon conversion and the nominal value

$$r_j = 20mm,$$

the predicted actuator-to-joint slope is approximately

$$\frac{0.3278°}{count}.$$

The model intentionally excludes hysteresis, relaxation, frictional memory, and related transmission nonlinearities. It therefore serves as the nominal sim-to-real baseline rather than as a complete physical model of the plant.

3.2 Direction-Bit Empirical Model

The quasi-static characterization revealed clear path dependence. As an initial empirical representation, the joint angle was modeled using the signed differential coordinate x together with a binary direction state $d \in \{-1,+1\}$:

$$q = 0.2782x - 0.297 - 0.878d.$$

This model achieved an RMSE of approximately 0.135° on the quasi-static characterization data. The direction-bit representation captures the separation between increasing and decreasing branches, but it has an important structural limitation: the memory state changes instantaneously when the direction of motion reverses. It therefore cannot represent partial reversal of the internal transmission state.

3.3 Model v1: Continuous Memory and Relaxation

To represent gradual rather than instantaneous memory reversal, Model v1 introduces a bounded internal state $z \in [-1,1]$. The memory-dependent equilibrium joint angle is

$$q_{eq} = ax + b + hz,$$

with identified parameters

$$a = \frac{0.2821°}{count}, \mathrm{b} = \text{-}0.284°, h = -0.956°.$$

The memory state evolves with differential displacement according to

$$z_{k+1} = d + (z_k - d)^{\frac{|\Delta x|}{L}}$$

where d = sign(Δx) and the identified transition scale is

$$\mathrm{L} = 3.310 \text{ counts.}$$

When the commanded differential coordinate is stationary, z is held rather than relaxing toward zero. Separate long-dwell measurements showed that the physical joint continues to move after the motors have reached their encoder targets. Joint angle is therefore modeled as relaxing toward q_eq with first-order dynamics using the representative time constant

$$\tau = 0.584 \text{ s.}$$

Model v1 is intended as a compact empirical stateful model of the observed behavior. The internal state z should not be interpreted as a directly identified physical friction variable or as a first-principles model of tendon mechanics.

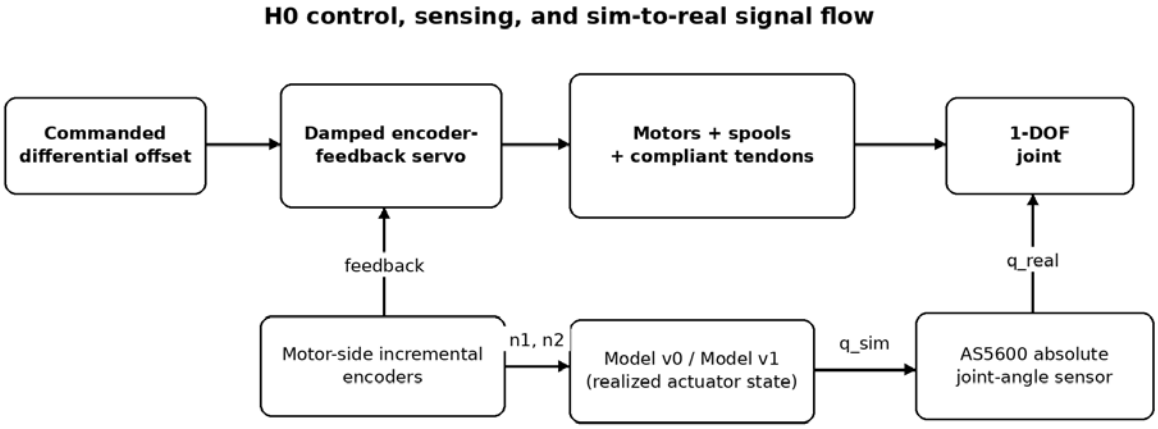


Figure 5. H0 control, sensing, and sim-to-real signal flow. The models are evaluated from realized motor-side actuator states and compared against the independently measured AS5600 joint angle.

4. Hardware Commissioning and Experimental Reference

Before characterization, the physical platform was commissioned so that motor-positioning errors would not dominate the later sim-to-real comparison. The final damped encoder-feedback servo reaches commanded winding positions without the large overshoot observed during early testing, allowing the measured motor-side encoder states to be treated as reliable experimental inputs rather than as an additional uncontrolled source of error.

4.1 Establishing the Experimental Reference

Because the motor encoders are incremental, they do not retain an absolute mechanical reference after the controller loses its state. A repeatable initialization procedure is therefore established at the beginning of each powered experimental session. The mechanism is manually placed in a calibrated geometric-zero configuration and the corresponding motor encoder states are defined as the reference positions $n_{1,0}$ and $n_{2,0}$. The AS5600 provides an independent check that the joint is at approximately q = 0.

Once this reference is established, the controller remains powered and connected so that the incremental motor-position reference is preserved. A scripted routine can return the motors to the stored baseline

before a run, after which the standardized common preload is applied. This procedure gives successive experiments a common actuator and joint starting condition within the session.

4.2 Tendon Preload

All final experiments use a standardized common preload of +100 motor winding counts on both actuators. Differential commands are then applied about this common operating point. The common preload maintains tension in both compliant tendon paths while allowing the signed differential coordinate x to be varied without changing the initialization procedure. The same preload convention is used during characterization, model identification, and final sim-to-real evaluation.

4.3 Limitation of the Current Referencing Procedure

The procedure provides a practical and repeatable reference for controlled experimental sessions, but it depends on preserving the incremental encoder state. A controller reset or power interruption removes the motor-side absolute reference and requires the calibrated configuration to be established manually again.

This limitation does not prevent the present study, because the plant can remain powered throughout a batch of trials. It does, however, motivate a future autonomous homing procedure capable of recovering the reference configuration without persistent incremental encoder counts. Such a procedure can use small controlled motor perturbations together with the resulting joint-angle response to infer the current tendon configuration and iteratively recover the calibrated state.

5. Characterization Methodology

The purpose of characterization is to determine how the physical compliant transmission maps known motor-side winding states to joint configuration before independent predictive tests are interpreted. All experiments begin from the calibrated reference and the +100-count common preload described in Chapter 4.

5.1 Quasi-Static Differential Sweep

The full quasi-static characterization uses the signed differential offset sequence

$$+5, +10, +15, +20, +25, +30, +25, +20, +15, +10, +5, 0,$$

followed by the symmetric negative excursion down to −30 counts and back to zero. A 5 s dwell is used at every measurement point before the joint angle is recorded. The trajectory intentionally revisits the same differential coordinates from different motion histories, allowing path dependence to be measured directly rather than inferred from separate experiments.

5.2 Hysteresis Characterization

Repeated coordinates exhibit clear branch separation. Depending on the coordinate, the measured difference between branches is approximately 1.4°–2.1°. Within an individual branch, however, the actuator-to-joint relationship remains highly linear. Identified branch slopes are approximately 0.252–0.286°/count, all below the nominal Model-v0 prediction of 0.3278°/count.

The quasi-static dataset is used for parameter identification. Consequently, later comparisons of Model v1 against this same dataset are treated as characterization/model-fit evidence rather than as independent predictive validation.

5.3 Relaxation Characterization

Long-dwell experiments are performed separately from the quasi-static sweep. These measurements show that joint motion continues after the motor encoders have already reached their targets. A first-order fit yields a representative relaxation time constant of $\tau = 0.584$ s.

Adding this first-order relaxation to the earlier direction-branch empirical model reduces trajectory RMSE on the relaxation probe from 0.290° to 0.191°, corresponding to a 34.1% improvement. This establishes that finite-time relaxation contributes measurably to the sim-to-real discrepancy.

6. Sim-to-Real Experimental Procedure

6.1 Matched Simulation and Hardware Conditions

For each selected operating point, the physical system is initialized using the same reference and preload procedure. The commanded motor motion is executed and the realized motor encoder states are recorded. These measured actuator states, rather than nominal command values alone, are supplied to the simulation models. The hardware output q_{real} is measured independently by the AS5600, while each model produces q_{sim} from the realized actuator state.

For each operating point, the sim-to-real discrepancy is defined as

$$e_q = q_{real} - q_{sim}.$$

This construction minimizes contamination of the comparison by residual motor-positioning error and isolates the mismatch associated with the compliant transmission and the assumptions of the model.

6.2 Identification and Validation Separation

The quasi-static characterization data are used for empirical model identification and therefore are not treated as independent predictive validation. Model v1 achieves approximately 0.115° RMSE on that identification dataset, compared with 0.135° for the direction-bit model.

After Model v1 is identified, two additional experiments are considered. First, a separate large-reversal relaxation probe tests whether the continuous-memory formulation improves dynamic prediction. Second, after the limitations of that probe become apparent, a dedicated minor-loop experiment is designed specifically to exercise intermediate memory states that were not present in the identification data. The minor-loop experiment is performed only after the Model-v1 parameters are fixed and constitutes the principal independent predictive validation in this study.

6.3 Targeted Minor-Loop Experiment

The minor-loop trajectory contains the partial reversals

$$+20 \rightarrow +18 \rightarrow +20$$

and

$$-20 \rightarrow -18 \rightarrow -20.$$

These 2-count reversals are deliberately smaller than the identified memory-transition scale $L = 3.310$ counts. Model v1 therefore predicts only partial movement of z toward the opposite branch, whereas the direction-bit model changes branch immediately. The experiment is thus specifically discriminative between instantaneous and continuous memory reversal.

7. Results and Discussion

7.1 Quasi-Static Sim-to-Real Behavior

The quasi-static experiment exposes two limitations of the nominal memoryless model. First, measured branch slopes of approximately 0.252–0.286°/count are lower than the Model-v0 prediction of 0.3278°/count. Second, repeated actuator coordinates produce different joint angles depending on the path by which the coordinate was reached, with branch separations of approximately 1.4°–2.1°.

Using the canonical quasi-static experiment schedule for both hardware and models, the final comparison gives

$$\text{Model v0 RMSE} = 1.364°$$

and

$$\text{Model v1 RMSE} = 0.107°,$$

corresponding to a 92.2% reduction in RMSE. Because the same quasi-static experiment supplies the information used for Model-v1 identification, this result is interpreted primarily as characterization/model-fit evidence rather than independent predictive generalization.

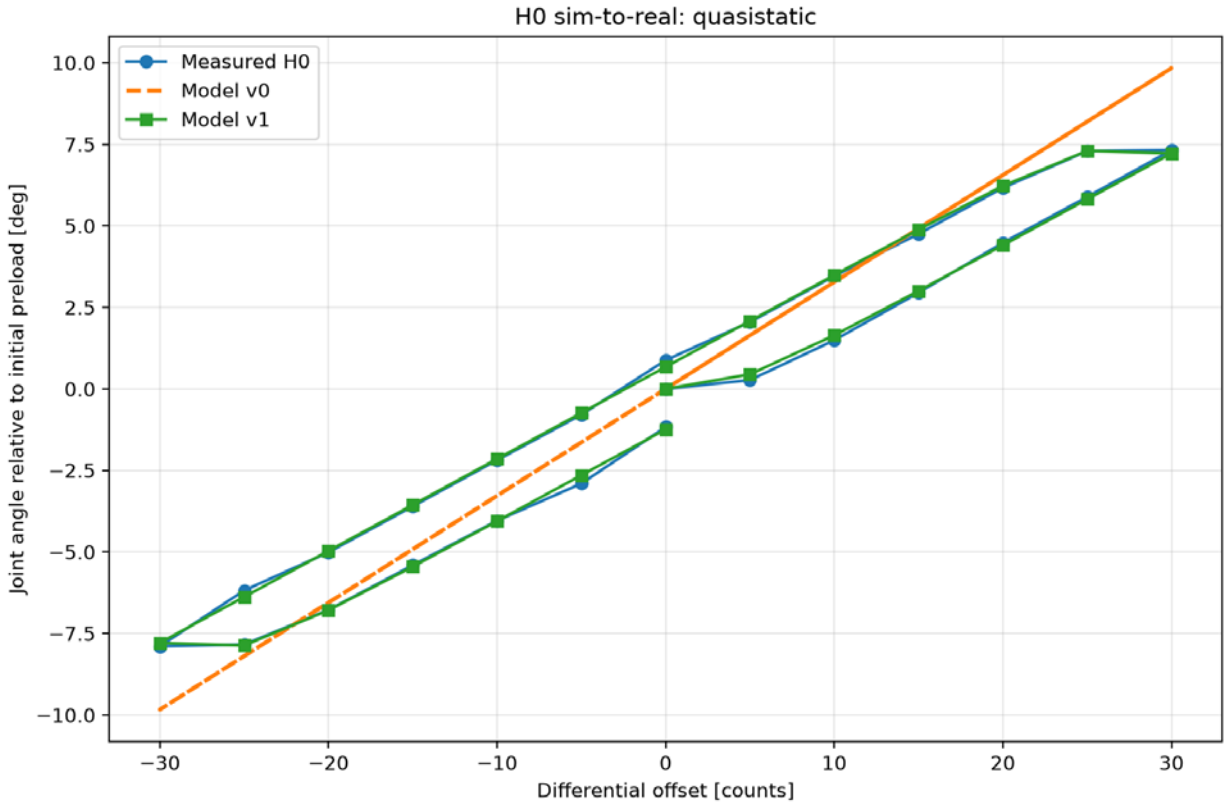


Figure 6. Quasi-static sim-to-real comparison in the differential-offset domain.

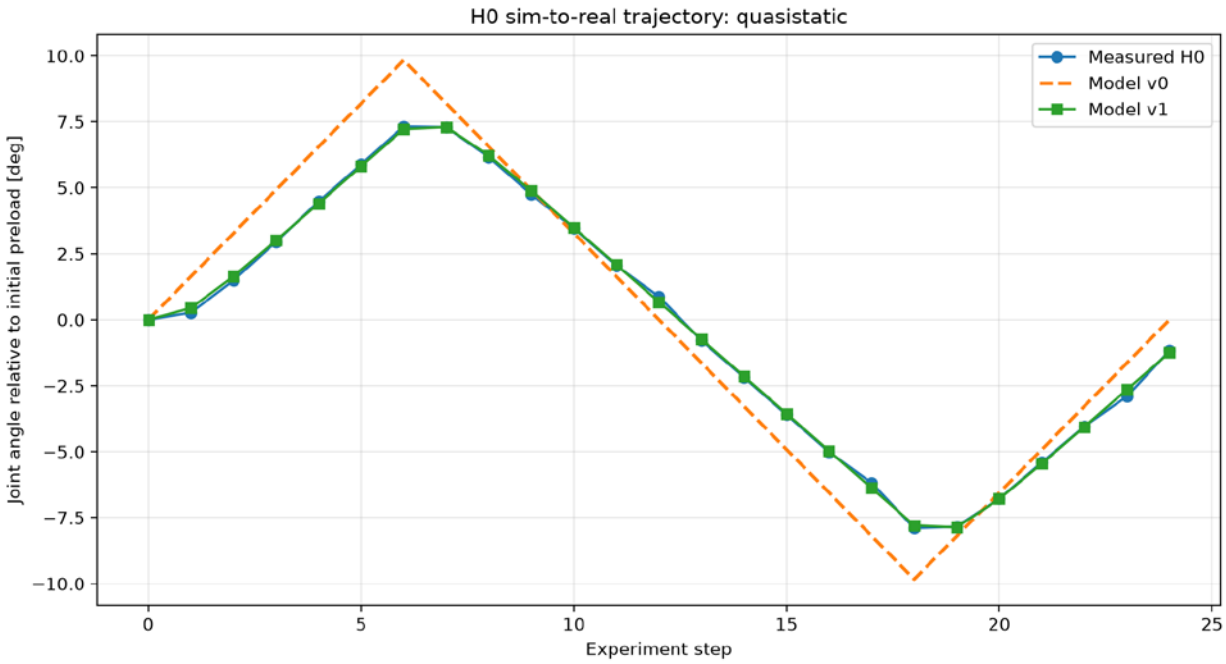


Figure 7. Chronological representation of the quasi-static characterization sequence.

7.2 Relaxation and the Negative Large-Reversal Result

The relaxation experiments demonstrate that quasi-static branch structure alone is insufficient to describe the system. Even after both motors have reached their encoder targets, the joint continues to creep toward its eventual configuration. Introducing first-order relaxation with $\tau = 0.584$ s reduces RMSE on the relaxation probe from 0.290° to 0.191°, confirming that the temporal response contributes measurably to the mismatch.

However, replacing the binary direction state with the continuous-memory state does not improve performance on the separate large-reversal relaxation probe. The direction-bit dynamic model achieves 0.191° RMSE, whereas Model v1 achieves 0.215° RMSE. Model v1 is therefore approximately 12.4% worse on this experiment.

This negative result is retained because it reveals an important limitation of the evidence available at that stage. A large reversal quickly drives the bounded memory state z toward an extreme, making the continuous-memory response increasingly similar to an instantaneous direction switch. The large-loop relaxation probe therefore provides little direct evidence that intermediate memory states are necessary. Rather than ignoring this outcome, it motivates an experiment designed specifically around the regime in which the two formulations make different predictions.

7.3 Independent Minor-Loop Validation

The targeted minor-loop experiment introduces only 2 counts of reversal, smaller than the identified transition scale of 3.310 counts. At +18 after reversal from +20, the measured joint angle is +4.352°. The direction-bit model predicts +5.589°, while Model v1 predicts +4.707° with $z \approx +0.091$.

Similarly, at −18 after reversal from −20, the measured joint angle is −5.272°. The direction-bit model predicts −6.183°, whereas Model v1 predicts −5.273° with $z \approx -0.093$.

Across the complete minor-loop experiment, the canonical independent sim-to-real comparison gives

$$\text{Model v0 RMSE} = 1.256°$$

and

$$\text{Model v1 RMSE} = 0.217°,$$

corresponding to an 82.8% reduction in RMSE. These data were not used for Model-v1 identification. The minor-loop experiment therefore provides the strongest predictive evidence in the present study that retaining a partial internal memory state is useful for describing the H0 transmission.

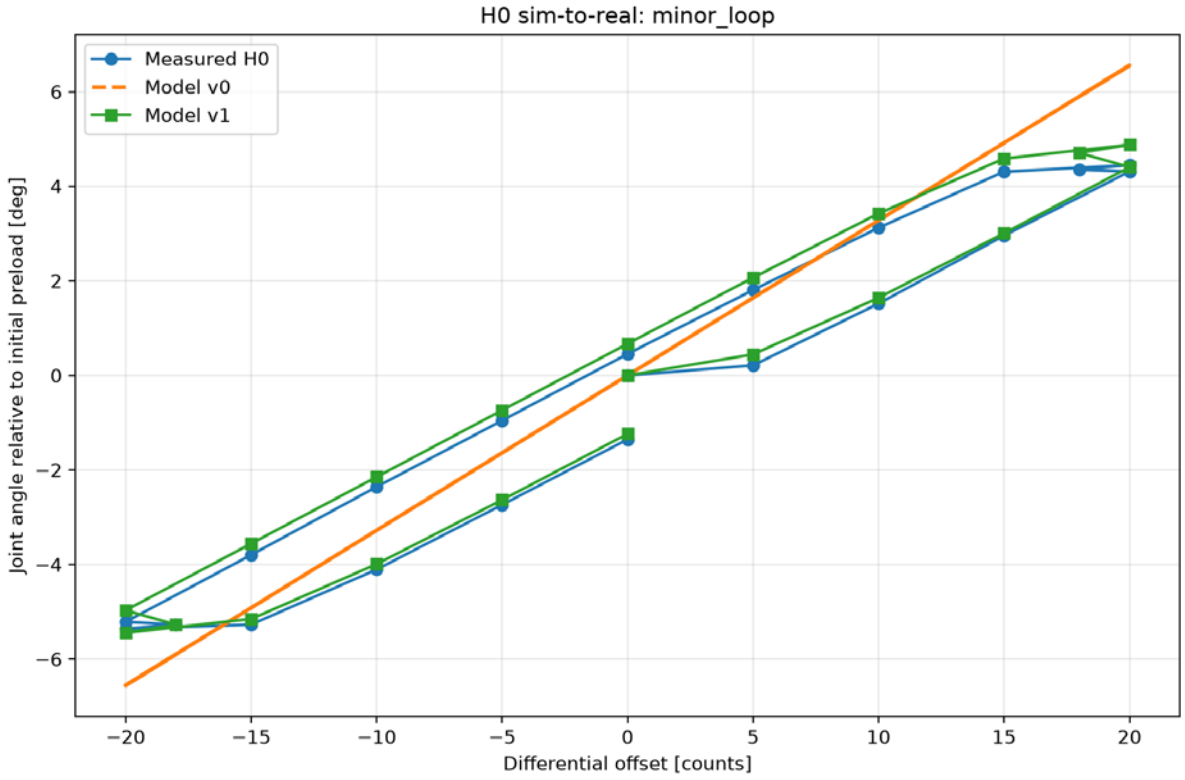


Figure 8. Independent minor-loop validation in the differential-offset domain.

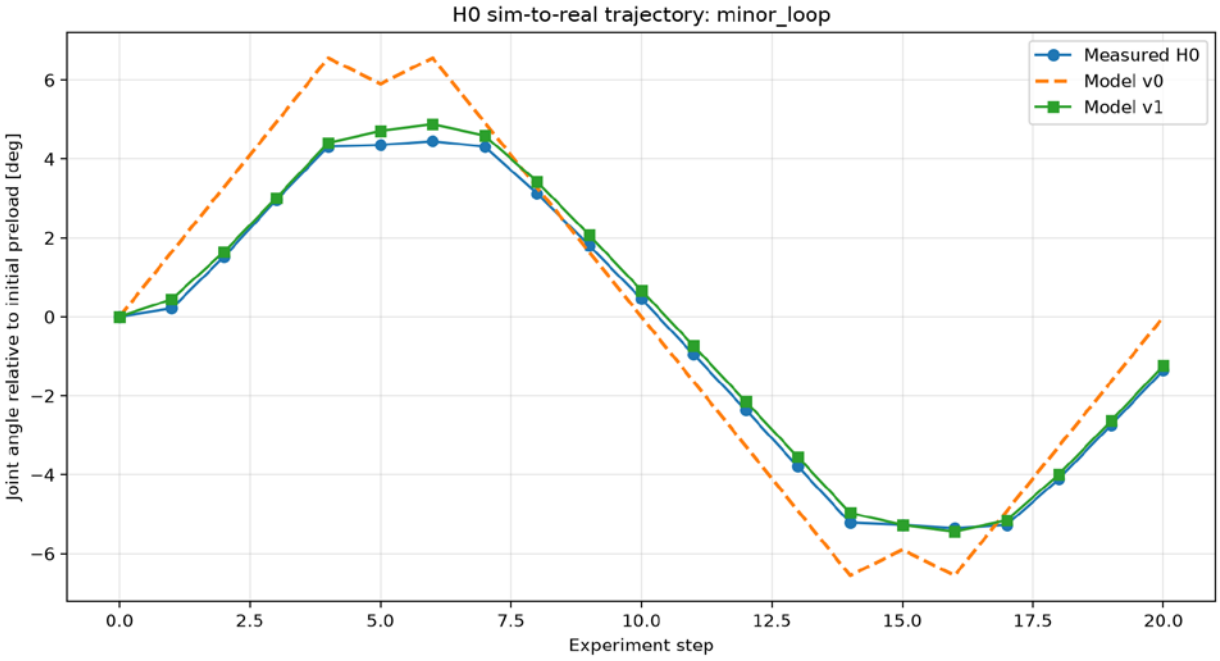


Figure 9. Chronological minor-loop validation.

Table 1. Compact summary of identification and validation results.

Experiment	Model / comparison	RMSE	Interpretation
Quasi-static identification	Direction-bit	0.135°	Identification fit
Quasi-static identification	Model v1	0.115°	Identification fit
Final quasi-static batch	Model v0	1.364°	Characterization baseline
Final quasi-static batch	Model v1	0.107°	92.2% reduction; not independent validation
Large-reversal relaxation probe	Direction-bit + dynamics	0.191°	Better on this probe
Large-reversal relaxation probe	Model v1	0.215°	12.4% worse
Independent minor loop	Model v0	1.256°	Predictive baseline
Independent minor loop	Model v1	0.217°	82.8% reduction

7.4 Interpretation

Taken together, the experiments indicate that the dominant sim-to-real discrepancy cannot be described solely as a constant scale error. The physical transmission exhibits both path-dependent equilibrium behavior and finite-time relaxation.

The direction-bit model demonstrates that a simple branch-dependent representation already explains much of the quasi-static hysteresis. Model v1 improves the representation on the identification data by replacing the instantaneous branch switch with a bounded continuous memory state, but it does not outperform the simpler model in every experiment.

The large-reversal relaxation result is therefore central to the interpretation rather than an inconvenient exception. Model v1 is justified not by universal superiority, but by the targeted minor-loop validation in which the distinction between instantaneous and partial memory reversal is actually observable. Accordingly, z is best interpreted as an experimentally useful latent state. The current experiments do not establish whether it corresponds uniquely to tendon friction, spring behavior, contact redistribution, structural compliance, or a combination of mechanisms.

8. Limitations and Future Work

The present models are identified and validated only over the tested H0 operating range and the standardized +100-count preload condition. Further experiments should determine whether the identified parameters a, h, L, and τ remain stable across preload, velocity, larger joint excursions, repeated cycling, and hardware reassembly.

A second direction is to investigate the physical origin of the identified memory state. Model v1 is deliberately phenomenological: its predictive success on the minor loop does not establish that the exponential displacement-domain state transition corresponds to a unique first-principles friction or tendon-mechanics law. More targeted measurements would be required before assigning z a physical interpretation.

The current experimental reference procedure is also session-dependent because the motor encoders are incremental. A controller reset or power interruption removes the motor-side reference and requires the geometric-zero configuration to be established manually again. Future work will therefore develop an autonomous homing procedure that can recover the calibrated configuration after reference loss. The intended approach uses small controlled actuator perturbations together with the resulting change in

joint angle to infer the current tendon/spool regime and iteratively return the system to the known baseline without adding dedicated force sensing.

Finally, the current work deliberately focuses on a minimal one-degree-of-freedom plant. Extending the model to more complex tendon routing, coupled joints, or broader dynamic operation should be treated as a later question rather than assumed from the H0 results.

9. Conclusion

This work developed and experimentally evaluated a one-degree-of-freedom antagonistic tendon-driven compliant joint intended as a minimal sim-to-real test platform.

The nominal memoryless physics-derived model predicts an actuator-to-joint slope of approximately 0.3278°/count, while physical characterization reveals lower branch slopes, substantial hysteresis, and time-dependent relaxation. These observations motivate progressively richer compact empirical models.

A binary direction-dependent representation captures the principal quasi-static branches, while long-dwell experiments identify a representative first-order relaxation time constant of 0.584 s. A continuous bounded memory state is subsequently introduced in Model v1 to represent partial hysteresis reversal.

Importantly, Model v1 does not outperform the simpler direction-bit model on a separate large-reversal relaxation probe. Rather than discarding this result, the experiment is recognized as insufficiently discriminative between the two memory formulations. This motivates a targeted minor-loop experiment using reversals smaller than the identified memory-transition scale.

On this independent validation experiment, Model v1 reduces RMSE from 1.256° for Model v0 to 0.217°, an 82.8% reduction. On the quasi-static characterization batch, Model v1 reduces RMSE from 1.364° to 0.107°, although this result is appropriately treated as model-fit evidence rather than independent validation.

The results support the use of a compact stateful empirical model for the H0 platform while leaving the physical origin and generality of the identified memory dynamics as open questions for future work.